NUMPY: TOTAL SALES AND PERCENTAGE INCREASE

AIM

To calculate the total sales for four quarters and determine the percentage increase in sales from the first quarter to the fourth quarter using NumPy.

ALGORITHM

1. Start.
2. Import the NumPy library.
3. Create a NumPy array containing the sales amounts for Q1, Q2, Q3, and Q4.
4. Calculate the total annual sales using np.sum().
5. Calculate the percentage increase using ((Q4 sales - Q1 sales) / Q1 sales) × 100.
6. Display the total sales and percentage increase.
7. Stop.

PROGRAM (PYTHON)

```
import numpy as np

# Sales for Q1, Q2, Q3, and Q4
sales_data = np.array([10000, 12000, 15000, 18000])

# Calculate total sales for the year
total_sales = np.sum(sales_data)

# Calculate percentage increase from Q1 to Q4
q1_sales = sales_data[0]
q4_sales = sales_data[3]

if q1_sales != 0:
    percentage_increase = ((q4_sales - q1_sales) / q1_sales) * 100
    print("Total annual sales:", total_sales)
    print("Percentage increase from Q1 to Q4:",
          round(percentage_increase, 2), "%")
else:
    print("Percentage increase cannot be calculated because Q1 sales are zero.")
```

INPUT

Quarterly sales data:

Quarter	Sales Amount

Q1	10000
Q2	12000
Q3	15000
Q4	18000

OUTPUT

```
Total annual sales: 55000
Percentage increase from Q1 to Q4: 80.0 %
```

RESULT

The total sales for the year are 55,000. Sales increased by 80% from the first quarter (10,000) to the fourth quarter (18,000).